TEXORA · LECTURE SHEET

Cutting Room Management

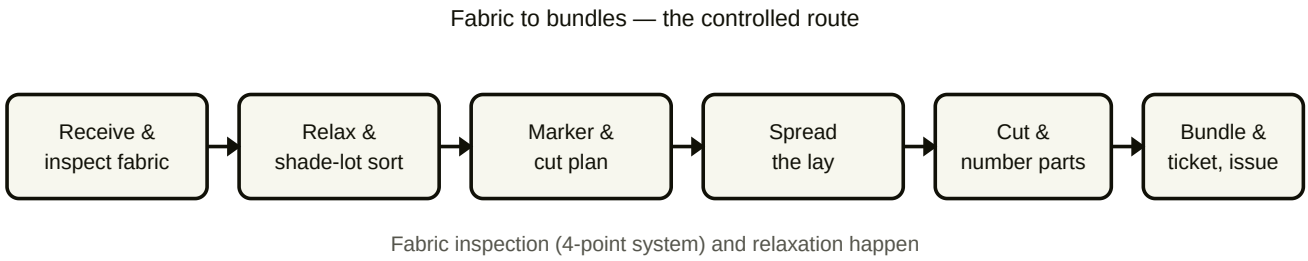
Managing the department where the money is: fabric reception and inspection, relaxation, marker planning, spreading, cutting and bundling — with marker efficiency and consumption arithmetic, and the wastage controls that protect 60–70% of the garment's cost.

AM-4201 · ADVANCED APPAREL MANUFACTURING-II · TOPIC 5 · LEVEL 4 TERM 2

1. Why the cutting room is managed hardest

Fabric is the largest single cost in a garment (Topic 1), and it becomes irreversibly committed in the cutting room: a sewing error costs minutes, a cutting error costs cloth. Cutting-room management therefore pursues three goals in order — **protect the material** (no metre consumed that the order did not budget), **protect quality** (accurate, shade-consistent, correctly identified parts), and **feed the lines** (bundles ready ahead of sewing, never the bottleneck, never flooding WIP). Its manager controls the order's profit more directly than any other department head.

2. The cutting-room workflow



before value is added — a defect cut into panels costs ten times its fabric value.Numbering ties every part to its lay and ply: shade matching at sewing and audit traceability both depend on it.

Figure 1 — The cutting-room route. Each arrow is also a record: inspection report, relaxation log, marker report, lay record, bundle tickets.

STAGE	MANAGEMENT CONTENT

Fabric inspection	Roll-by-roll check on the inspection machine; defects graded by the 4-point system; shade bands compared against the standard; width and length verified against the invoice — claims against the mill start here, with evidence.
Relaxation	Knits and elastane fabrics are unrolled and rested (commonly 12–24 h or per mill instruction) so spreading tension and roll memory release before cutting; cutting a tense knit guarantees garments that shrink out of tolerance after washing.
Marker planning	Pieces nested per ratio (sizes per marker) and fabric width, by CAD (AM-4102 Study 4) — the efficiency figure of §3 is set here, at a desk, before any cloth moves.
Spreading	Lays built ply by ply (manual, semi- or fully automatic spreaders); face direction per fabric (face-to-face for pairs/symmetry, face-one-way for nap/shade-critical cloth); ply height limited by cutting accuracy and fabric type.
Cutting	Straight knife for the lay, band knife/die for small parts, CNC cutters for precision at volume. Blade condition, lay compression and operator technique decide edge accuracy; notches and drills placed per the pattern.
Numbering & bundling	Every part numbered to its ply; parts sorted into operation bundles with tickets (style, size, shade lot, quantity, operations) — the sewing line's unit of work in PBS (Topic 2).

3. The arithmetic of consumption

WORKED EXAMPLES — VERIFY EVERY NUMBER

(a) Marker efficiency. Marker length 7.2 m, fabric width 1.45 m → marker area = 7.2 × 1.45 = 10.44 m². Pattern-piece area total = 8.62 m². Efficiency = 8.62 ÷ 10.44 × 100 = **82.6%** — the remaining 17.4% is marker waste, the price of geometry; every point of efficiency on a volume order is tonnes of cloth.

(b) Consumption per garment. The marker carries 6 garments → 7.2 ÷ 6 = 1.20 m/garment. Add cutting-room wastage allowance 3% (end losses, splices, defects cut around) → budgeted consumption = 1.20 × 1.03 = **1.236 m/garment**.

(c) Order fabric requirement. Order 12,000 garments × 1.236 = **14,832 m** to issue/buy against this marker and width. If the marker is replanned at a different ratio or width, consumption changes and the buy must change with it — purchasing from an old marker is a classic silent loss.

4. Wastage: where fabric actually leaks

WASTAGE SOURCE	CONTROL

Marker waste (geometry)	Better nesting, ratio optimisation, width selection per marker; CAD trials before commitment.
End loss (lay ends)	Marker length matched to lay length with minimum allowance; disciplined end cutting.
Splice loss & defects	Inspection data feeding splice planning; overlap allowances standardised, not operator-chosen.
Remnants & roll ends	Remnant inventory with lengths labelled; small orders and recuts draw from remnants first.
Cutting damage / recuts	Blade discipline, ply-height limits, first-piece checking against the pattern before the lay is fully cut.
Over-consumption drift	Per-order reconciliation: fabric issued vs garments cut vs consumption budget — variance investigated per order, not per season (Topic 6).

5. Quality and safety duties

Cut quality is checked as **first-piece approval** (top, middle and bottom ply measured against the pattern) and by bundle audits; shade continuity is protected by keeping one shade lot within a garment via numbering. Safety is non-negotiable around blades: chain-mesh gloves for knife operators, blade guards, no loose clothing, CNC safety interlocks — the cutting room holds the factory's sharpest hazards and its injury record is a management KPI alongside efficiency.

Glossary

TERM	MEANING
Marker	The arrangement of all pattern pieces for a lay, at a stated ratio and fabric width; its length sets consumption.
Lay	The stack of fabric plies spread to the marker length, cut in one operation.
Marker efficiency	Pattern-piece area ÷ marker area × 100.
4-point system	Fabric inspection grading: defects penalised 1–4 points by size; fabric rated in points per 100 square yards/metres.
Splice	Overlapping join where a defective section is cut out during spreading.

Key points and summary

- Cutting commits the costliest material irreversibly — management priority: protect material, protect quality, feed the lines.
- Consumption is decided at the marker desk: efficiency 82.6% and 1.236 m/garment in the worked standard.
- Wastage is itemised (marker, end, splice, remnant, damage) and reconciled per order.
- Inspection, relaxation and numbering are quality systems, not preliminaries.

Practice

Answer in your own words; brief model answers follow.

1. A marker is 8.4 m long on 1.50 m width and carries 7 garments; piece area totals 10.9 m². Calculate marker efficiency and consumption per garment.
2. Using (1), with a 3% wastage allowance, how much fabric must be bought for 9,800 garments?
3. Why is knit fabric relaxed before spreading, and what defect appears in finished garments if it is not?
4. State two jobs of the numbering system.
5. Why is first-piece checking done on top, middle and bottom plies rather than the top ply alone?

Answer key

1. Marker area = 8.4 × 1.50 = 12.6 m². Efficiency = 10.9 ÷ 12.6 × 100 = 86.5%. Consumption = 8.4 ÷ 7 = 1.20 m/garment.
2. Budgeted consumption = 1.20 × 1.03 = 1.236 m. Fabric = 9,800 × 1.236 = 12,112.8 m ≈ 12,113 m.
3. Rolling and winding hold the knit under tension and store elastic/spirality strain. Spread and cut in that state, the parts later recover — garments shrink and twist after making/washing, out of measurement tolerance. Relaxation lets the fabric reach its stable dimensions before it is measured into a marker.
4. (i) Shade/lot integrity: parts of one garment come from the same ply region, preventing shade mismatch within a garment. (ii) Traceability: any defect found later can be traced to lay and ply, and replacements are cut from the matching fabric position.
5. Knife deflection and lay compression grow with depth: bottom plies are where inaccuracy (smaller/larger parts, drifted notches) concentrates. A true top ply proves nothing about the bottom of the lay.